



ALIAH UNIVERSITY

Odd (Autumn) Semester Examination 2025-26

Paper Code: MBAUGHU01 ; Paper Name: Industrial Economics & Management

Programme: B.TECH VII Semester, 4TH YEAR

Full Marks: 80; Time: 3 Hours.

(The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as possible)

GROUP: A (Answer all the questions)

(1 × 10 = 10)

1. Economics is primarily concerned with:

- (a) Unlimited wants & unlimited resources
- (b) Limited wants & unlimited resources
- (c) Unlimited wants & limited resources
- (d) Limited wants & limited resources

2. Demand law explains the relationship between:

- (a) Price and supply
- (b) Price and demand
- (c) Income and demand
- (d) Demand and cost

3. Which of the following is a part of organizational behaviour?

- (a) Consumer behaviour
- (b) Machine behaviour
- (c) Employee motivation
- (d) Natural behaviour

4. Maslow's theory is related to:

- (a) Cost reduction
- (b) Hierarchy of needs
- (c) Product quality
- (d) Market segmentation

5. Production management mainly deals with:

- (a) Finance and profit
- (b) Buying of goods
- (c) Conversion of raw materials into finished products
- (d) Packaging only

6. The main aim of Total Quality Management (TQM) is:

- (a) Minimum output and maximum cost
- (b) Zero defect and continuous improvement
- (c) Customer avoidance
- (d) Increase waste

7. Which tool is used for quality control?

- (a) Control charts
- (b) Balance sheet
- (c) Marketing mix

(d) Job evaluation

8. Fixed capital refers to:

- (a) Wages paid to labour
- (b) Investment in machinery and buildings
- (c) Transport cost
- (d) Short-term loans

9. The recruitment process in HRM is concerned with:

- (a) Terminating employees
- (b) Selecting the wrong candidates
- (c) Hiring and attracting eligible candidates
- (d) Producing goods

10. 4Ps of marketing include:

- (a) People, Price, Product, Profit
- (b) Product, Price, Place, Promotion
- (c) Policy, Promotion, Process, People
- (d) Plan, Price, People, Place

GROUP: B (Answer any five questions)

(5 × 5 = 25)

1. Define economics and explain its importance in industry.
2. What are the different types of market structures? (Explain any four)
3. Explain motivation and its importance in organizational behavior.
4. What are the main functions of production management?
5. Define quality and explain the concept of TQM.
6. What are the objectives of financial management?
7. Explain the various sources of recruitment in HRM.
8. What is marketing mix? Explain its elements.

GROUP: C (Answer any three questions)

(15 × 3 = 45)

1. Discuss the law of supply and demand with diagrams and examples. Explain its application in industrial decision-making.
2. Explain the major theories of motivation and their importance in organizational behavior.
3. Describe the stages involved in production planning and control. How does it help in effective manufacturing?
4. Discuss the various financial statements used in financial management and their significance in business decision making.

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Autumn Semester Examination - 2025

B.Tech 4th year, 7th semester Examination

Paper Name: Machine Learning and Soft Computing

Paper Code: CSEUGPC24

Full Marks: 80

Time: 3 hrs

Group-A (Answer all questions)		(2x5=10)
1	What is Address Biasness?	CO1 – Und 2
2	Differentiate the text-based recognition system and topic modelling techniques in Big Data analytics.	CO5 – Ana 2
3	Differentiate the Latent Dirichlet Allocation (LDA) and Latent Semantic Analysis (LSA) topic modelling techniques.	CO5 – Ana 2
4	Distinguish between fitness-proportionate selection (roulette wheel) and rank-based selection in GA reproduction.	CO3 – Ana 2
5	What is overfitting in MLPs? Suggest methods to reduce it.	CO4 – Und 2
Group-B (Answer any four questions)		5 x 4=20
6	Perform derivations for the margin and hard-SVM equations mathematically.	CO4 – App 5
7	Write an algorithm for SGD (Stochastic Gradient Descent) for a Neural Network.	CO4 – Dev 5
8	Explain a kernel function and an activation function in machine learning, and give suitable examples of each.	CO4 – App 5
9	Write short notes on activation functions:(i) Rank Selection operation, (ii) Multi-point crossover, (iii) L2 & L1-Norm.	CO3 – Und 5
10	Give a brief explanation of the conceptual working flow diagram of a fuzzy-based system.	CO2 – App 5
Group-C (Answer any five questions)		5x10=50
11	Maximize the function $f(x) = 6x^2 - 5x + 7$, where $x = 4, 9, 14$, with (Chromosome size = 6 such that (i) selection operation (Rank Selection), (ii) Uniform crossover, (iii) Up to two iterations. How do different feature engineering strategies impact model accuracy and training efficiency in high-dimensional datasets? If removing noisy features reduces the error rate from 20% to 12%, by how many percentage points is the error reduced? Compute softmax for values (1, 2, 3).	CO3 – App, CO5 – Ana, CO5 – App, CO4 – App 5+2+2+1=10

12	Analyze the Training and Testing operation of a Recurrent Neural Network. What are the steps involved in designing a fuzzy logic controller? Explain the different types of membership functions used in the fuzzification process. List and explain the main phases of NLP preprocessing from raw text to feature extraction.	CO4 – Ana, CO2 – App, CO1 – Und, CO4 – Und 2+2+3+3=10
13	Explain the terms “Chromosome, Gene, Allele, Locus, Genotype, Phenotype” with proper examples for a GA problem. Provide the parameter requirements for effective clustering algorithms. What are Core and Support in fuzzy logic?	CO3 – Und, CO5 – Ana, CO1 – Und 5+3+2=10
14	Consider the following two fuzzy sets A and B defined over a universe of discourse [0,5] of real numbers with their membership function: $\mu_A(x) = \frac{x}{1+x}$ and $\mu_B(x) = 2^{-x}$. Determine the membership functions of (i) $\bar{A}, \bar{B}$ and (ii) $A \cap B$ and draw them graphically.	CO2 – App, CO1 – Und 4+6=10
15	Suppose a genetic algorithm uses chromosomes of the form $x = abcdefgh$ with a fixed length of eight genes. Each gene can be any digit between 0 and 9. Let the fitness of individual x be calculated as: $f(x) = (a + b) * (c + d) + (e + f) - (g + h)$. Let the initial population consist of four individuals with the following chromosomes: $x_1 = 7\ 2\ 4\ 1\ 3\ 5\ 3\ 2$; $x_2 = 9\ 7\ 1\ 2\ 1\ 6\ 0\ 1$; $x_3 = 5\ 3\ 2\ 2\ 1\ 2\ 8\ 5$; $x_4 = 7\ 1\ 8\ 5\ 2\ 4\ 9\ 4$. Use the following (i) Evaluate the fitness of each individual, (ii) Cross the fittest two individuals using one-point crossover at the middle point, (iii) Evaluate the fitness of the new population with the best four chromosomes (two-old and two-new) (iv) Perform (ii) to (iii) up to three iterations.	CO3 – Dev 10
16	Establish a minimum distance classifier (MDC) for a 3-class classification problem. How do different model evaluation protocols, such as leave-one-out, k-fold cross-validation, confusion-matrix analysis, and N-K train-test-test splits, impact the reliability of a machine-learning pipeline from raw data preprocessing to final model performance for Big Data analytics?	CO4 – App CO5 – Ana 4+6=10

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AUTUMN SEMESTER EXAMINATION 2025
FOR BTECH 4TH YEAR / 7TH SEM ONLY

Paper Name: Professional Elective III (Computer Vision)
Paper Code: CSEUGPE15

Full Marks: 80
Time: 3:00 Hrs

Section A
(Answer all questions)

5 × 2

1. (a) What is the size of an image of size 32x32 having 16 gray levels?
(b) What is meant by hyperparameter?
(c) What is the time complexity of nearest neighbor classifier for training and prediction?
(d) What is meant by style transfer?
(e) 10 filters of size 5x5 are convolved on a data volume of 32x32x3 with stride 1 and pad 2. What will be the size of the output volume?

Section B
(Answer any 6 questions)

6 × 5

2. Briefly mention how computer vision may be useful in the medical field. Also mention other application areas of computer vision. 2+3
3. How is a digital image represented? Mention the three image acquisition modalities. Which one is the most prevalent now-a-days? 2+2+1
4. Give the first order and the second order derivative of an image. Also indicate how they may be useful. 1+2+2
5. Briefly mention the present-day challenges in computer vision. 5
6. What is feature vector? How is it useful in object recognition? 2+3
7. Differentiate between face detection and face recognition. How is a hyperplane represented? 3+2
8. What is meant by a receptive field? How is convolution related to a neuron? 2+3
9. State the ReLU activation function and its significance. Explain the max-pool operation with an example. 3+2

Section C
(Answer any 4 questions)

4 × 10

10. (a) With a suitable diagram, explain the steps of a traditional 8+2

computer vision system.

(b) What is the use of a depth-camera?

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|-----|--|-----|
| 11. | (a) Define Laplacian of Gaussian. | 4+6 |
| | (b) How is it useful in edge detection? | |
| 12. | (a) What is connected component labelling? | 6+4 |
| | (b) How is it useful in feature extraction? | |
| 13. | (a) Briefly explain how a linear classifier works. | 6+4 |
| | (b) What are the limitations of a linear classifier? | |
| 14. | (a) Show the basic building block of a Convolutional Neural Network. | 7+3 |
| | (b) State the use and significance of softmax activation function. | |
| 15. | Write short notes. | 5+5 |
| | (a) Visual data | |
| | (b) Deep learning in computer vision | |

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End-Semester Examination - 2025

(For BTech CSE 7th Semester)

Paper Name: *Natural Language Processing*

Paper Code: *CSEUGPE11 [Professional Elective-II]*

Full Marks: 80

Time: 3 Hrs

Group – A

(5X2=10)

1. Answer the following questions –

- State the role of regular expression in NLP. (Und, CO2, CO1)
- Mention some commonly used word embedding techniques. (Und, CO2)
- State the theorem used in Naïve-Bayes classifier. (Rem, CO3)
- What is meant by 'stopwords'? Mention some stopwords in English. (Rem, CO2)
- What is Name-Entity Recognition? Give examples to show its importance. (Ana, CO2)

Group – B

(Answer any 5 questions)

(5X6=30)

- Briefly explain various levels of Natural Language Processing. (Und, CO1, CO2) 6
- What is word tokenization? Discuss various issues to perform word tokenization. (Und, CO2, CO1) 2+4
- Write a note on N-gram model. (Und, CO3, CO1) 6
- Discuss some commonly used text normalization techniques. Give examples. (Und, CO2) 6
- Define Sentiment Analysis. What evaluation metrics are used for text classification? Illustrate the metrics precision, recall, and F1-score with the formulas. (Und, CO5) 1+2+3
- Construct Deterministic Finite Automata to accept all the strings of the language – (App, CO1) 6
 $L: \text{All strings ending with } ba, \text{ where } \Sigma = \{a, b\}$

Group – C

(Answer any 4 questions)

(10X4=40)

- Differentiate between Stemming and Lemmatization with their use cases. (App, CO4) 6
 - Mention some major steps of Porter Stemmer algorithm used for stemming. (Und, CO1, CO2) 4
- Explain Noisy-channel model and mention how it is used for spelling correction. (Und, CO4) 6
 - Define 'derivation' in grammars. How LMD and RMD concludes whether a Context Free Grammar is ambiguous or not? Give examples. (Und, CO5) 4
- Write minimum edit distance algorithm (use Levenstein weight for operations). (Und, CO1) 4
 - Calculate minimum edit distance for two strings **SERVICE** and **SEVERE**. Mention any one alignment for your answer. (App, CO1) 6
- What is the need of smoothing? Explain Good-Turing Discounting algorithm used for smoothing. (Und, CO3) 1+3

b) Answer the following questions based on the corpus given below-

<s> *There is a big garden* </s>

<s> *Children play in a garden* </s>

<s> *They play inside beautiful garden* </s>

Using bi-gram probabilistic model calculate the probability for the sentence:

<s> *They play in a big garden* </s>

Also calculate the probability of the same using Add-1 smoothing in the same model.

3+3

(Ana, CO3)

12. A speech-based medical monitoring system uses a Hidden Markov Model (HMM) to determine a person's health condition from observable symptoms. The model consists of two hidden health states and three possible observable symptoms. The parameters of the HMM are provided below.

Hidden states – H – means Healthy, and F – means Fever.

Observable states – n – means normal, c – means cold, and d – means dizzy.

Initial probability π , Transition Matrix A and Emission/ Observation Matrix B is given as followings -

$$\pi = \begin{bmatrix} \pi(H) \\ \pi(F) \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix} \quad A = \begin{bmatrix} P(H \rightarrow H) & P(H \rightarrow F) \\ P(F \rightarrow H) & P(F \rightarrow F) \end{bmatrix} = \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$$

$$B = \begin{bmatrix} P(n|H) & P(c|H) & P(d|H) \\ P(n|F) & P(c|F) & P(d|F) \end{bmatrix} = \begin{bmatrix} 0.5 & 0.4 & 0.1 \\ 0.1 & 0.3 & 0.6 \end{bmatrix}$$

Over three consecutive days, the system records the following symptoms:

$O = [\text{normal (n), cold (c), dizzy (d)}]$

Calculate the sequence of health condition for the above-mentioned dates using Viterbi Algorithm.

(App, CO4, CO5) 10

13. a) Differentiate between Top-Down and Bottom-Up parsing. (Und, CO3)

4

- b) Compare and contrast among document-level, sentence-level, and aspect-level sentiment analysis. (Ana, CO4, CO5)

6
